

Daily Geometry Drill #4

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Exterior angles sum to 360° · Isosceles chains · Regular polygons as equal triangles · Similar triangles · Area ratios.

Attempt each question before checking answers. Time yourself. No calculators. Diagrams are not to scale.

Section A Rapid Recognition

(≤15 s each)

Exterior angles of any polygon sum to 360° ; interior = $180^\circ - \text{exterior}$. Know these cold.

- Write down the size of each interior angle of a regular 12-sided polygon.
- An isosceles triangle has apex angle 40° . Write down each base angle.
- The incircle of the right triangle with legs 3 and 4 touches all three sides. What is the inradius?
- Each interior angle of a regular polygon is 162° . Write down the number of sides.
- Write down the number of diagonals of a decagon.
- The angle bisector from A of triangle ABC meets BC at D , with $AB : AC = 2 : 3$ and $BC = 10$. Write down DC .
- The isosceles triangle with sides 5, 5, 6 has base 6. Write down the median from the apex to the base.
- A median of a triangle has length 12. How far is the centroid from the vertex at the end of that median?
- Two similar triangles have corresponding sides in the ratio $2 : 3$. The smaller has area 8. Write down the area of the larger.
- Write down the area of a regular hexagon with side 2.

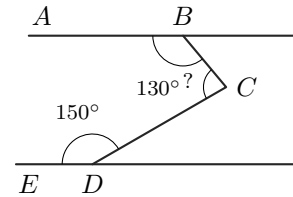
Section B Manipulation Drills

(≤50 s each)

Chase angles through parallels, isosceles triangles and regular polygons. Draw the extra line.

1. AB is parallel to ED . Given $\angle ABC = 130^\circ$ and $\angle CDE = 150^\circ$, find $\angle BCD$.

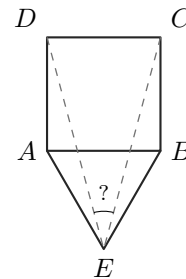
A) 60°
 B) 70°
 C) 80°
 D) 100°
 E) 110°



2. $ABCD$ is a square and ABE is an equilateral triangle drawn outside the square on side AB . What is $\angle DEC$?

(after Primer 1.4 P11)

A) 15°
 B) 20°
 C) 45°
 D) 60°
 E) 30°



3. The triangle with sides 13, 14, 15 has area 84. Its inscribed circle has what radius?
4. The five diagonals of a regular pentagon form a five-pointed star whose points are the pentagon's vertices. What is the angle at each point of the star?

A) 36°
 B) 30°
 C) 45°
 D) 54°
 E) 72°

5. $ABCDEF$ is a regular hexagon. What fraction of its area is triangle ACE ?

A) $\frac{1}{3}$
 B) $\frac{1}{2}$
 C) $\frac{2}{3}$
 D) $\frac{3}{4}$
 E) $\frac{1}{4}$

6. A square, a regular hexagon and one other regular polygon fit together around a point with no gaps or overlaps. How many sides does the other polygon have?

A) 8
 B) 10
 C) 15
 D) 12

- E) 18
7. Each interior angle of a regular polygon is four times each exterior angle. How many sides does it have?
- A) 10
B) 8
C) 9
D) 12
E) 16
8. $ABCDEFGH$ is a regular octagon. Find $\angle ACF$.
9. A regular pentagon and a regular hexagon share a side and lie on opposite sides of it. At each end of the shared side there is a gap between the two shapes. What is the angle of this gap?
- A) 120°
B) 126°
C) 128°
D) 132°
E) 144°
10. A five-pointed star is drawn with five straight strokes, not necessarily regular. Write down the sum of the angles at its five points.

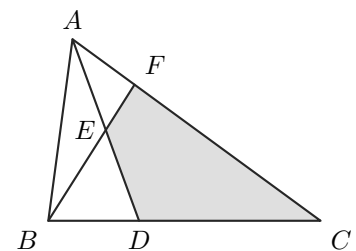
Section C Substitution & Structure

(≤ 90 s each)

SMC Q18–22 level. Split the figure into pieces you know (equilateral triangles, isosceles triangles, similar copies), then use ratios.

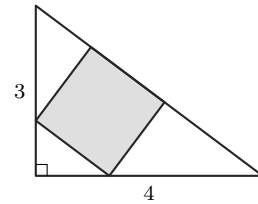
1. In triangle ABC , the point D lies on BC with $BD : DC = 1 : 2$, and E is the midpoint of AD . The line BE meets AC at F . What fraction of triangle ABC is the shaded quadrilateral $EDCF$?

- A) $\frac{1}{2}$
B) $\frac{5}{8}$
C) $\frac{7}{12}$
D) $\frac{2}{3}$
E) $\frac{3}{4}$



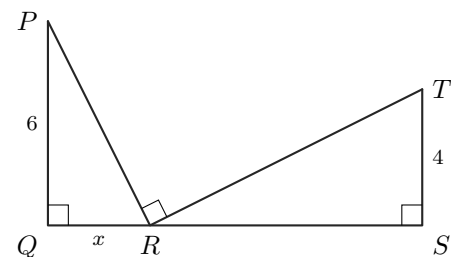
2. A right-angled triangle has legs 3 and 4. A square sits inside it with one side along the hypotenuse and its other two vertices on the legs. What is the side length of the square?

- A) $\frac{12}{7}$
 B) $\frac{5}{3}$
 C) $\frac{3}{2}$
 D) $\frac{60}{37}$
 E) $\frac{60}{49}$



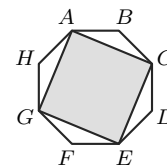
3. PQ and TS are perpendicular to QS , with $PQ = 6$, $TS = 4$ and $QS = 11$. The point R lies on QS with $QR = x$, and $\angle PRT = 90^\circ$. Then (after Primer 1.4 P48)

- A) x has two possible values whose sum is 12
 B) x has two possible values whose difference is 5
 C) x has only one value, and $x \geq 5.5$
 D) x has only one value, and $x < 5.5$
 E) x cannot be determined from the given information



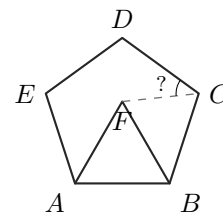
4. $ABCDEFGH$ is a regular octagon with sides of length 1. What is the area of the shaded square $ACEG$?

- A) $2 + \sqrt{2}$
 B) $1 + \sqrt{2}$
 C) $2 + 2\sqrt{2}$
 D) 4
 E) $3 + 2\sqrt{2}$



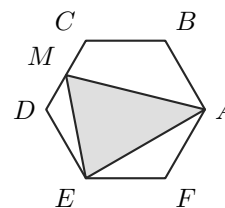
5. $ABCDE$ is a regular pentagon, and ABF is an equilateral triangle with F inside the pentagon. What is $\angle FCD$?

- A) 36°
 B) 48°
 C) 42°
 D) 54°
 E) 66°



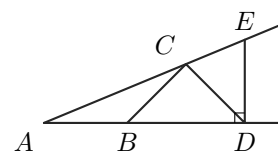
6. $ABCDEF$ is a regular hexagon and M is the midpoint of CD . What fraction of the hexagon is the shaded triangle AME ?

- A) $\frac{1}{3}$
 B) $\frac{1}{2}$
 C) $\frac{1}{4}$
 D) $\frac{3}{8}$
 E) $\frac{5}{12}$



7. The points B and D lie on one arm of an angle at A , and C and E lie on the other arm, with $AB = BC = CD = DE$. The line DE is perpendicular to AD . What is $\angle BAC$?

- A) 15°
 B) 18°
 C) 20°
 D) 22.5°
 E) 30°



8. A regular 12-sided polygon is inscribed in a circle of radius 1. What is the total area inside the circle but outside the polygon?

- A) $\pi - 2$
 B) $\pi - 3$
 C) $\pi - \frac{3\sqrt{3}}{2}$
 D) $\pi - 2\sqrt{2}$
 E) $\frac{\pi}{12}$

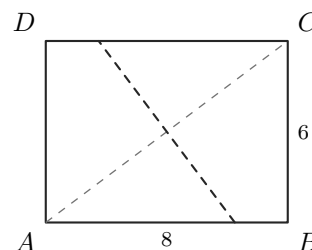
Section D Challenge

(SMC Q21–25 / BMO1 difficulty)

The key step is a construction, a hidden equal length or a similar pair. Diagrams are not to scale.

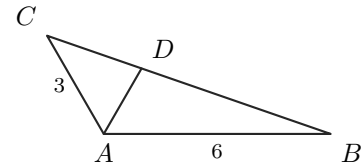
1. A rectangular sheet $ABCD$ has $AB = 8$ and $BC = 6$. It is folded along a straight crease so that C lands exactly on A . How long is the crease?

- A) 7
 B) $5\sqrt{2}$
 C) $\frac{25}{4}$
 D) 8
 E) $\frac{15}{2}$



2. In triangle ABC , $AB = 6$, $AC = 3$ and $\angle BAC = 120^\circ$. The bisector of angle A meets BC at D . What is the length of AD ?

A) $2\sqrt{3}$
 B) 2
 C) 4
 D) $\sqrt{7}$
 E) 3

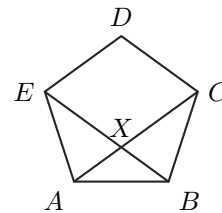


3. What is the side length of the largest square that fits inside a regular hexagon with sides of length 1?

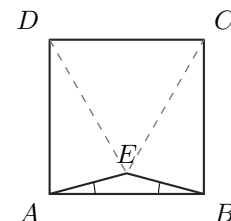
A) 1
 B) $\frac{2\sqrt{3}}{3}$
 C) $3 - \sqrt{3}$
 D) $\frac{\sqrt{6}}{2}$
 E) $\sqrt{2}$

4. $ABCDE$ is a regular pentagon with sides of length 1. The diagonals AC and BE meet at X . What is the length AX ?

A) $\frac{\sqrt{5} - 1}{2}$
 B) $\frac{\sqrt{5} + 1}{2}$
 C) $\frac{1}{2}$
 D) $\frac{\sqrt{5}}{2}$
 E) $\frac{3 - \sqrt{5}}{2}$



5. $ABCD$ is a square. The point E inside the square satisfies $\angle EAB = \angle EBA = 15^\circ$. Prove that triangle CDE is equilateral.



“Every polygon is a bundle of triangles: count the exterior angles, find the isosceles pairs, and let similar copies carry the ratios.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths